

A decorative graphic consisting of thin, dark gray lines that resemble a circuit board or a network diagram. These lines extend horizontally from the left and right edges of the central black box, with several small circles at the end of the lines, suggesting nodes or connections.

BOOTSTRAP-BASED ASSESSMENT OF UNCERTAINTY IN LOGISTIC REGRESSION FOR BREAST CANCER DIAGNOSIS

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PROJECT GOAL

- Evaluate logistic regression on WDBC dataset
- Identify stable predictors of malignancy
- Apply 3 bootstrap methods (2000 resamples each)
- Compare bootstrap vs Wald inference
- Assess coefficient stability and uncertainty

WHY BOOTSTRAPPING?

- Logistic regression relies on large-sample assumptions
- Biomedical predictors are highly correlated
- Wald intervals may underestimate variability
- Bootstrapping is data-driven and assumption-light



NONPARAMETRIC BOOTSTRAP

- Resample rows with replacement
 - Very flexible
 - Minimal assumptions
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PARAMETRIC BOOTSTRAP

- Assume logistic model is correct
- Simulate new outcomes from fitted probabilities
- Predictors remain fixed

RESIDUAL-BASED BOOTSTRAP

- Resample deviance residuals
- Add to linear predictor
- Inverse logit $\rightarrow$ simulate new responses

SIGNIFICANT PREDICTORS

- mean_texture
- mean_smoothness
- mean_concave_points
- Consistently significant across all bootstrap methods.

UNSTABLE PREDICTORS

- mean_radius
 - mean_perimeter
 - mean_compactness
 - mean_concavity
 - mean_symmetry
 - mean_fractal_dimension
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- Wide intervals overlapped zero → unreliable effects.

BOOTSTRAP VS WALD INFERENCE

- Bootstrap SEs matched Wald SEs
- Bootstrap CIs similar to Wald CIs
- No underestimation of variability
- Logistic regression assumptions were adequate

CONCLUSION

- Bootstrapping validated logistic regression reliability
- Only 3 predictors were stable & reproducible
- Others unstable due to multicollinearity
- Resampling improves transparency of uncertainty